Numerical treatment of Schrodinger's equation

Solving the Schrodinger equation numerically

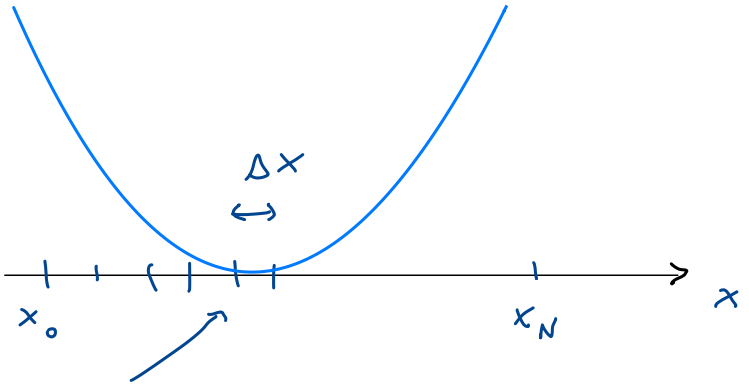
consider 1D problem

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \underbrace{\frac{\alpha}{4}x^4}_{V(x)} \qquad \text{e.g.}$$

Task: find eigenstates + energies E

$$(*) \quad \left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right)\varphi_E(x) = E\,\varphi_E(x)$$

Strategy I: discretization



$$x_i = x_0 + i\,\Delta x \qquad i = 0, \dots, N$$

$$\frac{\partial^2\varphi}{\partial x^2}(x_i) \approx \frac{1}{2}\frac{1}{\Delta x^2}\left(\varphi(x_{i+1}) + \varphi(x_{i-1}) - 2\varphi(x_i)\right)$$

$\varphi(x_{i+1}) = \varphi_{i+1}$, $\varphi(x_{i-1}) = \varphi_{i-1}$, $\varphi(x_i) = \varphi_i$

$$(*) \Rightarrow \quad -\underbrace{\frac{\hbar^2}{4m\,\Delta x^2}}_{=:\,\kappa}\left(\varphi_{i+1} + \varphi_{i-1} - 2\varphi_i\right) + V_i\,\varphi_i = E\,\varphi_i$$

$$\begin{bmatrix} \ddots & & & \\ -\mathcal{H} & 2\cdot\mathcal{H}+V_9 & -\mathcal{H} & \\ & -\mathcal{H} & 2\cdot\mathcal{H}+V_{10} & -\mathcal{H} \\ & & -\mathcal{H} & \ddots \end{bmatrix} \begin{bmatrix} \varphi_0 \\ \vdots \\ \varphi_9 \\ \varphi_{10} \\ \varphi_{11} \end{bmatrix} = E \begin{bmatrix} \varphi_0 \\ \vdots \\ \varphi_{10} \end{bmatrix}$$

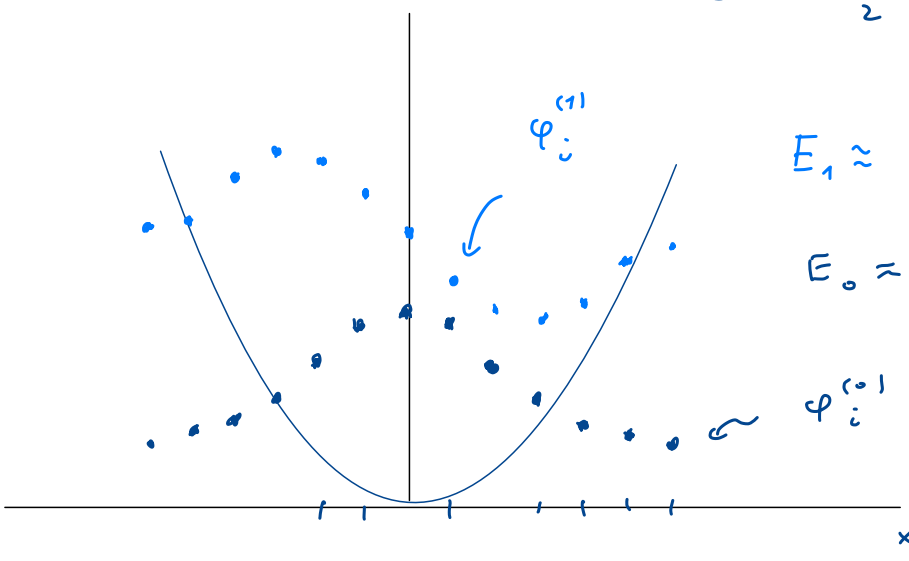
$\uparrow$ 9 $\quad\uparrow$ 10 $\quad\uparrow$ 11 $\quad\uparrow$ Hermitian matrix

$\Rightarrow$ solve eigenproblem numerically

$\Rightarrow$ approximate values of E, $\varphi_E(i)$

$\downarrow$

$$V(x) = \frac{m\omega^2}{2} x^2$$



$E_1 \approx \hbar\omega \cdot 3/2$

$E_0 \approx \hbar\omega/2$

Strategy II — variational method

Statement: ground state wave function minimizes $\langle\psi|\hat{H}|\psi\rangle$ with $\langle\psi|\psi\rangle = 1$

Alternative formulation:

$$E_{\text{ground state}} = \min_{\psi} \frac{\langle\psi|\hat{H}|\psi\rangle}{\langle\psi|\psi\rangle}$$

Consider family of functions: $|\psi(p_1, p_2 \ldots)\rangle$ (parameters)

$$\text{minimize } \frac{\langle\psi|\hat{H}|\psi\rangle}{\langle\psi|\psi\rangle} = E(p_1, p_2, \ldots)$$

simple example:

$$H = \underbrace{-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}}_{\hat{T}} + \underbrace{\frac{\alpha}{4}x^4}_{\hat{V}}$$

dimensionless units $\frac{\hbar^2}{m}\frac{1}{a_0^2} = \alpha a_0^4$ $\Rightarrow$ length $a_0 \equiv \left(\frac{\hbar^2}{m\alpha}\right)^{1/6}$;

energy $\epsilon_0 \equiv \alpha\left(\frac{\hbar^2}{m\cdot\alpha}\right)^{2/3}$

$$\Rightarrow \quad \tilde{H} \equiv H/\epsilon_0 = -\frac{1}{2}\frac{\partial^2}{\partial\xi^2} + \frac{1}{4}\xi^4 \qquad \xi \equiv \frac{x}{a_0}$$

Try $\psi = e^{-\xi^2/2p}$

- $\langle\psi|\psi\rangle = \int d\xi\ e^{-\xi^2/p} = \sqrt{\pi\cdot p}$
- $\langle\psi|\xi^4|\psi\rangle = p^{5/2}\int dy\ e^{-y^2}y^4 = p^{5/2}\frac{3}{4}\sqrt{\pi}$
- $\langle\psi|-\frac{\partial^2}{\partial\xi^2}\psi\rangle = \ldots = \frac{1}{2}\sqrt{\frac{\pi}{p}}$

$$\frac{\langle\psi|\tilde{H}|\psi\rangle}{\langle\psi|\psi\rangle} = \frac{1}{4}\frac{1}{b} + \frac{3}{16}\cdot b^2$$

b_{min}: $\Rightarrow \quad b_{min} = \left(\frac{2}{3}\right)^{1/3}$

$$E_0 \approx \epsilon_0 \cdot \frac{3}{8}\cdot\left(\frac{3}{2}\right)^{1/3} = 0{,}423\ \epsilon_0$$

Strategy II/b

consider a "nice" set of functions (basis)

for example, Harmonic oscillator wave functions

$$\varphi_n(x) \sim H_n\left(\frac{x}{a}\right) e^{-x^2/2a^2}$$

(free to choose a ...)

Properties:

- normalized: $\int_{-\infty}^{\infty} \varphi_n(x)^2\, dx = 1$
- orthogonal: $\int_{-\infty}^{\infty} dx\ \varphi_n^*(x)\,\varphi_m(x) = 0 \quad$ for $n \neq m$
- complete: "any" $\psi(x)$ can be expressed as

$$\psi(x) = \sum_{n=0}^{\infty} a_n \cdot \varphi_n(x)$$

Variational Ansatz

$$\psi(x) = \sum_{n=0}^{N} a_n \cdot \varphi_n(x)$$

Recipie

1 Compute matrix elements $n, m = 1, \dots, N$

$$H_{mn} \equiv \int dx\ \varphi_m^*(x)\ \hat{H}\ \varphi_n(x) =$$

$$= \int dx\ \varphi_m(x) \left(-\frac{\hbar^2}{2m} \frac{\partial^2 \varphi_n}{\partial x^2} + \frac{\alpha}{4} x^4\ \varphi_n(x) \right)$$

evaluate numerically

2 Construct matrix equation

$$\sum_{n=0}^{N} H_{mn}\ a_n = E\ a_m$$

$$\underline{\underline{H}} \cdot \underline{a} = E\ \underline{a}$$

3 find eigenvalues E of $\underline{\underline{H}}$ and corresponding coefficients $\{a_n^E\}$

4 Eigenstates are approximated as

$E \to$ $$\varphi_E(x) \equiv \sum_{n=0}^{N} a_n^E\ \varphi_n(x)$$

5 Normalization $$\sum_{n=0}^{N} |a_n|^2 = 1$$

Some details

Then $\langle\psi|\hat{H}|\psi\rangle = \sum_{n,m=0}^{N} \alpha_n^* \langle\varphi_n|\hat{H}|\varphi_m\rangle \alpha_m$

$\sum_n \alpha_n \varphi_n$ $\quad$ $\sum_m \alpha_m \varphi_m$ $\quad$ H_{nm}

H_{nm} can be computed numerically

$$\langle\psi|\psi\rangle = \sum_{n,m=0}^{N} \alpha_n^* \underbrace{\langle\varphi_n|\varphi_m\rangle}_{\delta_{n,m}} \alpha_m = \sum_{n=0}^{N} |\alpha_n|^2$$

Task: minimize $\langle\psi|\hat{H}|\psi\rangle$ under condition

$\langle\psi|\psi\rangle = 1$,

i.e. minimize $\sum_{n,m=0}^{N} \alpha_n^* H_{nm} \alpha_m$ under condition $\sum_{n=0}^{N} \alpha_n^* \alpha_n = 1$

Solution with Lagrange method:

minimize $F(\alpha_n, \lambda) = \sum_{n,m=0}^{N} \alpha_n^* H_{nm} \alpha_m - \lambda\left(\sum_n \alpha_n^* \alpha_n - 1\right)$

some algebra $\Rightarrow$ $\sum_{m=0}^{N} H_{nm} \alpha_m = \lambda \cdot \alpha_n$

$\rightsquigarrow \{\alpha_0, \dots, \alpha_N\}$ is an eigenvector of truncated matrix

$$\begin{bmatrix} H_{00} & H_{01} & \dots & H_{0N} \\ H_{10} & H_{11} & \dots & H_{1N} \\ \vdots & & & \vdots \\ H_{N0} & \dots & & H_{NN} \end{bmatrix}$$

with eigenvalue λ_{min} $\Rightarrow$ $E_{min} = \lambda_{min} = \dfrac{\langle\psi|\hat{H}|\psi\rangle}{\langle\psi|\psi\rangle}$

This generalizes to other states!